

Semester: July 2025– November 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code: 03	Class: TY	Semester: V (SVU 2023)
Programme: B.Tech EXTC ENGG.		
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 216U03E513	Name of the Course: Database Management System	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	
i)	List and describe different types of database systems .	05
ii)	Define functional dependency. State and explain Armstrong's axioms	05
iii)	Discuss integrity and security features in SQL such as triggers, domain constraints, and authorization	05
Q.2	Attempt any one	10
i)	Consider the following library database schema: student(stud_id, name) course(course_id, title, instructor) enrolled(stud_id, course_id) a. Change the name of student with ID 'S201' to 'Neha Kapoor'. b. Find the total number of courses taken by students. c. Find the student(s) who have enrolled in at least one course taught by every instructor who teaches the course titled 'Database Systems'.	
ii)	Consider the following description of a Hospital Management System: A hospital has several departments (e.g., Cardiology, Neurology, Pediatrics). Each department is identified by a Department_ID and has a Department_Name and Location. Each department employs several doctors, each identified by a Doctor_ID. For each doctor, the hospital maintains their Name, Specialization, and Contact_No. Each patient is identified by a Patient_ID, and has attributes Name, Age, and Address. A patient can be treated by multiple doctors, and a doctor can treat multiple patients. Each treatment is recorded with Date, Diagnosis, and Prescription details. Patients are also admitted to rooms, each identified by a Room_No, with attributes Type (e.g., ICU, General) and Charges_per_day.	



	a) Identify all entities, attributes, and relationships from the above description. b) Draw a neat and well-labeled E-R Diagram showing entities, attributes, keys, and relationships. c) Mention any assumptions or cardinalities	
Q.3	Attempt any one	10
i)	Explain CAP theorem and its significance in distributed databases	
ii)	Compare Key-Value and Document Data Models, Column-Family Stores for NoSQL database.	
Q.4	Attempt the following Consider the relation schema: $R = (\text{EmpID}, \text{EmpName}, \text{DeptID}, \text{DeptName}, \text{ProjectID}, \text{ProjectName}, \text{HoursWorked})$ which represents employees working on various projects within departments. The following functional dependencies (FDs) are given: 1. $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$ 2. $\text{DeptID} \rightarrow \text{DeptName}$ 3. $\text{EmpID}, \text{ProjectID} \rightarrow \text{HoursWorked}$ 4. $\text{ProjectID} \rightarrow \text{ProjectName}$ a) Convert the relation R to First Normal Form (1NF). b) Decompose R into relations that are in Second Normal Form (2NF).	10
Q.5	Attempt the following Explain dense and sparse indices with examples. Compare ordered indexing and hashing.	10